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An Implantable Biohybrid Neural Interface Toward Synaptic Deep Brain Stimulation

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In patients with sensory nerve loss, such as those experiencing optic nerve damage that leads to vision loss, the thalamus no longer receives the corresponding sensory input. To restore functional sensory input, it is necessary to bypass the damaged circuits, which can be achieved by directly stimulating the appropriate sensory thalamic nuclei. However, available deep brain stimulation electrodes do not provide the resolution required for effective sensory restoration. Therefore, this work develops an implantable biohybrid neural interface aimed at innervating and synaptically stimulating deep brain targets. The interface combines a stretchable stimulation array with an aligned microfluidic axon guidance system seeded with neural spheroids to facilitate the development of a 3 mm long nerve-like structure. A bioresorbable hydrogel nerve conduit is used as a bridge between the tissue and the biohybrid implant. Stimulation of the spheroids within the biohybrid structure in vitro and use of high-density CMOS microelectrode arrays show faithful activity conduction across the device. Although functional in vivo innervation and synapse formation has not yet been achieved in this study, implantation of the biohybrid nerve onto the mouse cortex shows that neural spheroids grow axons in vivo and remain functionally active for more than 22 days post-implantation.

1. Introduction

The ability to regrow sensory nerves to the original destinations in the adult mammalian brain is restricted[1] due to the growth-inhibitory environment of the extracellular matrix in the central nervous system, which prevents axonal regeneration.[2,3] Optogenetic,[4–6] electrical,[7,8] ultrasound[9] and transcranial magnetic stimulation techniques[10] can be used to modulate neural activity in the brain. For sensory restoration applications in which the sensory nerves are damaged, stimulation interfaces primarily target the cortex due to its easy accessibility. Using dynamic current steering cortical implants have enabled patients to distinguish objects and letters using a very low number of electrodes.[11] More recent intracortical electrodes improved stimulation threshold and spatial resolution down to 400 µm.[12] However, it remains to be tested if future cortical implants will be able to deliver useful sensations. The biggest challenge preventing current implants from restoring natural-like sensation is the functional complexity and very large number of neurons involved in sensory processing. Restoring sensation at its original resolution requires gaining modulatory control of a very high number of neurons at an almost single-neuron resolution with cellular specificity which is impossible using standard stimulation electrodes. Therefore even a dramatic increase in channel count would not automatically improve the resolution of, e.g., visual perception[11,13] Stimulating the primary target regions of sensory organs could lead to a more natural perception of

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information, since the higher level sensory processing performed in the cortex is maintained and the lower number of involved neurons could potentially make it technically more feasible.[14,15] Targeting these regions is not without challenges: 1) they are often located in deep brain regions making access difficult and 2) the dense packing of neurons necessitates high specificity for stimulation targets.[16] Commercially available deep brain electrodes have up to eight electrodes that each stimulate a cubic millimeter of neural tissue including thousands of neurons[17] and axons. Reducing electrode size down to 5 m or implementing novel design strategies have increased the specificity to ~10 neurons;[8,18,19] nevertheless, the inability to precisely target an electrode to a single neuron,[20] lack of cell specificity,[21–24] and the immune response[25] limit the feasibility of this technology for long term sensory neurorehabilitation.

Biohybrid neural interfaces combine currently available stimulation methods with living neural cells to overcome limitations of traditional neural implants.[26–31] By embedding neurons into engineered scaffolds connected to high-resolution electrode arrays, neurons act as biological synaptic relays between implanted electronics and host neural tissue.[29] There are two main approaches to biohybrid interfaces. 1) One involves cell-seeded probes, where neurons or neural stem cells are cultured directly onto electrode arrays before implantation, improving biocompatibility and tissue integration[25,32] A more recent approach cultured muscle cells on a peripheral nerve multielectrode array to attract peripheral nerve axons into the implant and amplify the signal-to-noise ratio of recorded spikes.[30] 2) The second approach involves integrated neurons encouraged to grow out of the device into host tissue, effectively acting as a synaptic relay. This method addresses stiffness mismatch and biocompatibility issues by promoting natural integration between implanted cells and host tissue, reducing immune response and inflammation.[25] For example, Adewole et al. developed micro-tissue engineered neural networks (microTENNs), consisting of bidirectional neural aggregates with long axonal tracts encapsulated within a 3D scaffold designed for optical stimulation.[33,34] Despite their potential advantages such as higher resolution and biocompatibility, biohybrid neural interfaces are still very under-explored. Current challenges are ensuring long-term stability of implanted cells, controlling the activity of individual neurons on the implant, directing axons toward the appropriate target location and successful axonal integration and synapse formation of neurons extending axons from a biohybrid device into the host brain tissue.

While allogeneic and xenogeneic transplantations of neurons and organoids into adult brains have functionally integrated new neurons into developed neural networks,[35,36] neural implants have yet to incorporate this for improving biocompatibility, cell specificity and stimulation resolution. The integration of living neurons into biohybrid neural interfaces offers three key advantages: 1) it allows precise control over which neurons, and how many, are being stimulated by positioning neurons on the electrodes while providing functional and electrical insulation; 2) it exploits biological properties such as the ability to genetically modify neurons or to synaptically target axons from the implant to specific neural populations in the brain (e.g., excitatory versus inhibitory neurons); and 3) it enables axons to integrate into the neural tissue, forming connections that extend and cover the tissue in three dimensions. In this work, we present an implantable biohybrid neural interface aimed at targetting deep brain structures through synaptic stimulation (**Figure 1a**). A polydimethylsiloxane (PDMS) axon guidance structure containing spheroids of neurons is integrated with a stretchable stimulation array to enable isolated stimulation of individual spheroids. The axons grow in the microchannels and are gradually and directionally merged to form a nerve-like fascicle. This fascicle can reach a length of more than 3 mm as it is growing in a conduit (either PDMS-based or gelatin-based). The conduit is then the only part of the device to be implanted in vivo to lead the artificial nerve-like fascicle into deep brain regions. Besides demonstrating the full functionality of the biohybrid implant in vitro, we also show the feasibility of implantation and the survival and axonal growth of neurons in the hybrid implant in vivo.

In summary, we have made several advancements toward a functional biohybrid neural interface. The presented platform integrates microfluidic axon guidance structures onto a stretchable multi-electrode stimulation array, allowing precise guidance of neuronal growth with multiple input channels to stimulate a nerve-like structure. We developed a protocol to reliably grow nerve-like structures over 3 mm long within a fully PDMS-based implant. Additionally, we demonstrated the direct fabrication of hydrogel conduits onto PDMS microstructures with the goal to support axonal integration with the host tissue. Finally, we confirmed the feasibility of guided axonal growth within the implant after in vivo implantation, promoting the potential of biohybrid approaches for future synaptic stimulation. As an outlook, specific recommendations for future studies to achieve axonal integration and synaptic stimulation for fully functional biohybrid stimulation electrodes are discussed.

2. Results

2.1. Design of the Implantable Biohybrid Neural Interface

The concept of the presented biohybrid nerve model involves stimulating individual neural spheroids on a stretchable multielectrode array where each spheroid is structurally insulated from the others using PDMS microstructures. The axons of each spheroid are guided to form a more than 3 mm long nerve-like fascicle, which can be directed toward specific target structures within the brain. This configuration holds the potential to transform electrical stimulation into synaptic stimulation (Figure 1a). The biohybrid device consists of an axon guidance structure which is plasma bonded onto stretchable PDMS-based platinum multielectrode array (MEA). The MEA is connected to an omnetics connector for interfacing with external electronics (Figure 1b). The guidance structure contains 16 isolated neural seeding wells and microchannels converging all axons into a common nerve-forming structure that can be implanted and directed toward the targeted deep brain region. The electrodes sitting underneath the seeding wells enable stimulation of individual neural spheroids, while the channels constrain the growth of axons and therefore signal propagation from the seeding wells toward an implantable nerve conduit. We developed two implantable biohybrid neural interfaces: In the first implant the nerve-forming structure is based on a 3 mm long PDMS nerve channel that is monolithically integrated to the guidance structure and enables the formation of the artificial nerve-like structure (Figure 1c,e). The

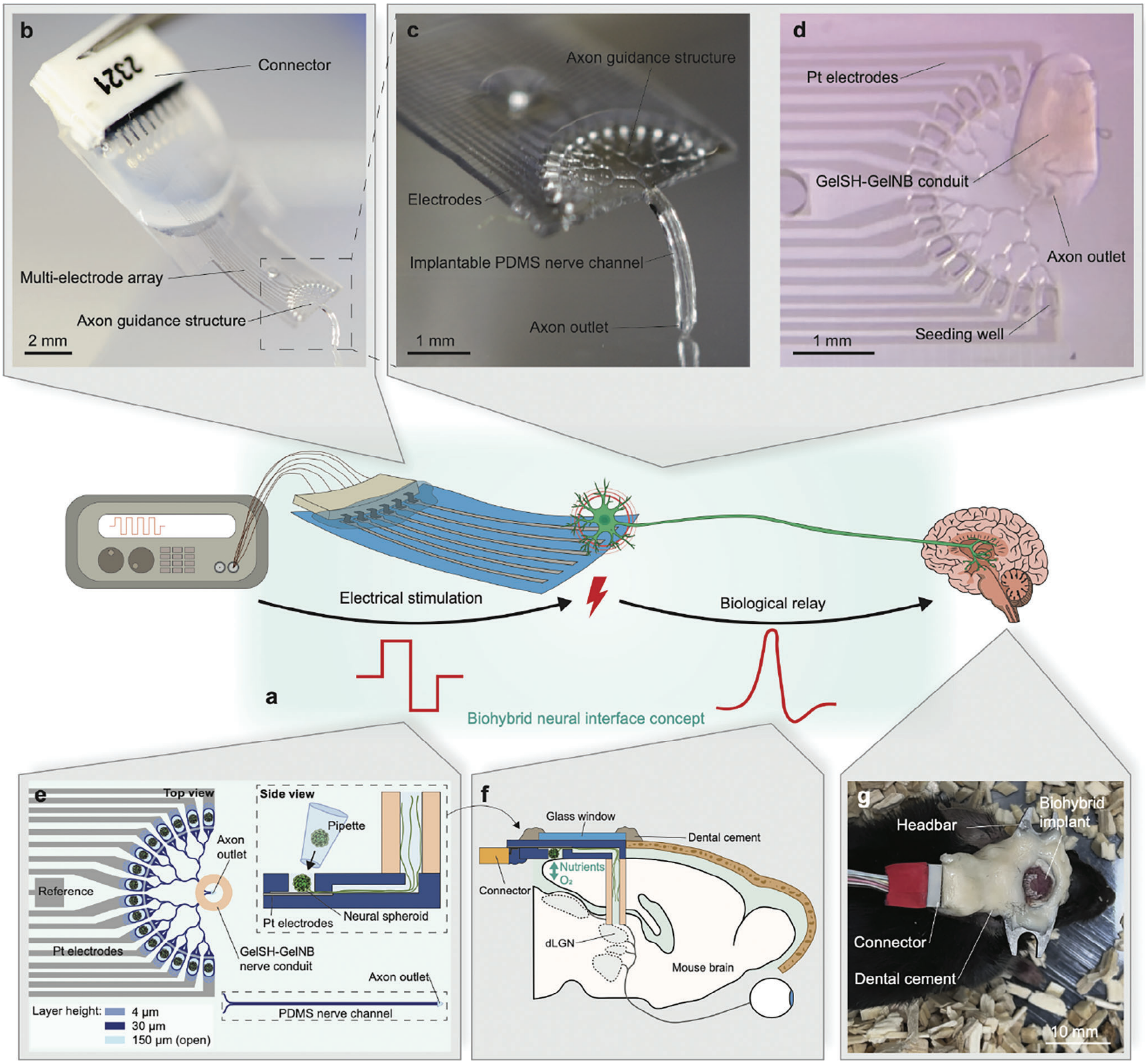

Figure 1. Concept and implementation of the implantable biohybrid neural interface. a) Schematic showing our implementation of the biohybrid concept. The neurons on the device act as biological relay to transmit a stimulation pulse to the brain. b) The implantable biohybrid neural interface with a PDMS based nerve channel. c) Magnification of the biohybrid axon guidance structure that is implanted on the cortex. The flexible PDMS nerve channel can be implanted into the brain to target the visual thalamus. d) Biohybrid implant in which the PDMS channel has been replaced with a GelSH-GelNB hydrogel conduit. e) Schematic of the biohybrid implant. The microfluidic axon guidance structure is aligned on top of the Pt-based microelectrode array and seeded with neural spheroids. The implantable nerve channel is either made of PDMS with an axon outlet at the end of the nerve/forming structure or consists of a GelSH-GelNB based hydrogel conduit vertically fabricated on top of the triangular axon outlet. Neural spheroids are seeded into each of the 16 seeding wells to grow axons toward the nerve channel. f) The biohybrid implant is flipped around and implanted onto the cortical surface of a craniotomized mouse brain. The PDMS or hydrogel nerve channel is directed toward the lateral geniculate nucleus (LGN). g) The implant is fixed onto the mouse head using a glass window and dental cement. A headbar enables subsequent mounting on a two-photon microscope for imaging axonal growth *in vivo as well as for calcium activity recordings.* During stimulation experiments the implant is connected through an omnetics connector.

conduit lies on the same plane as the microstructure and is bent into the brain when implanted while the rest of the device sits on the surface of the brain. In the second model, a UV-crosslinkable and bioresorbable GelSH-GelNB hydrogel conduit printed out of plane on the guidance structure replaces the PDMS nerve channel for increased biocompatibility to converge and direct axons toward their target region (Figure 1d,e).

Prior to implantation, neural spheroids (DIV1, from primary rat retinas or cortex) are prepared within a spheroid forming dish and seeded in each of the 16, 150 m deep, seeding wells on the following day (Figure 1e). On the same day of spheroid seeding the biohybrid implant is flipped upside down onto the craniotomized cortical surface such that the neural spheroids receive nutrients and oxygen directly from the brain (Figure 1e,f). To prevent spheroids from falling out, the implants are submersed in a highly viscous medium containing 1% methylcellulose. Neurons in the implant can grow axons along the microfluidic guidance system toward a target structure in the brain (Figure 1f). In our initial in vivo experiments the implant is fixed onto the mouse brain together with a metal headbar for two-photon imaging, which allows for both tracking of axonal growth in vivo as well as calcium activity recordings of spheroids of the implanted biohybrid device. For electrical stimulation, the implant is connected to external stimulation electronics via a multichannel omnetics connector (Figure 1g).

2.2. Fabrication and Characterization of the Biohybrid MEA

The implant has three layers: a PDMS substrate, microstructured tracks and electrodes, and a PDMS microstructure for electrical insulation and axonal guidance (**Figure 2**a; Figure S1, Supporting Information). Several design considerations were crucial for the successful growth of the nerve-like bundle. To ensure implantability onto the mouse cortex, the device's overall width was limited to 4 mm. Seeding wells with a diameter of 200 m were designed, matching the smallest electrode width we could comfortably fabricate. Each well accommodated up to 1000-cell spheroids[37] which we knew from previous studies should result in a sufficient amount of axons to form an artificial nerve like fascicle. The wells were arranged in a half-circle so that axons from each spheroid had the same distance to the nerve channel. The nerve channel was dimensioned to theoretically accommodate over 100 times the expected axon output from each spheroid (assuming an axon diameter of 0.3 m). This provided sufficient room for nutrient diffusion. The channel extended approximately 3.5 mm to reach the visual thalamus. Shallow merging angles directed axons toward the nerve channel. Hydrogel conduits with a 300 m inner diameter were used to account for alignment variations during fabrication onto the axon outlet. The tracks consist of a stack of SiO_2/Ti/Pt transferred on the PDMS substrate using template stripping transfer printing.[38] The tracks and electrodes have a specific meander shape on the micrometer scale[39] to improve mechanical compliance (Figure S2, Supporting Information). This microstructuring ensures that the electrodes won't break if subjected to moderate mechanical strain which could happen during the implantation procedure or while on the brain. The PDMS microstructure covers the electrode tracks up to the electrode sites assuring electric insulation. Seeding wells sit on top of the electrode with axon guidance channels leading to the nerve channel (Figure 2b). The microstructuring of the electrode also enables functional calcium imaging of the underlying spheroids (Figure 2c), allowing us to test the device by looking at the calcium response of neural cells behind the tracks both in vitro and in vivo. Since the fabrication process and device architecture is new and hasn't been reported before, extensive characterization was performed.

The electrochemical performance of the electrodes was assessed in phosphate-buffered saline solution (PBS) (see Methods). Cyclic voltammetry (CV) was performed (Figure 2d). The shape of the CV is as expected for smooth Pt microelectrodes,[40] and the electrochemical window was determined to be between -0.6 and +0.8 V (versus Ag/AgCl). The calculated cathodic charge storage capacity (CSCc) based on the CV is 1.02 ± 0.42 mC cm^{-2} (n = 13). According to electrochemical impedance spectroscopy (EIS) the microelectrodes exhibited an impedance of 8 ± 1.9 kΩ (n = 13) at 1kHz Figure 2e (the surface area of one electrode is 34373 μm^2). The current injection performance of the electrodes was then evaluated. As can be seen in Figure 2f, the current at which the electrode polarization surpasses the electrochemical window occurs between the injection of 50 and 75 µA. This corresponds to a charge injection capacity (CIC) of 50 ± 14 µC cm^{-2} (n = 13). Next, we investigated the effect of seeding neural spheroids on the electrodes. We compared the electrochemical performance of in vitro (neurobasal media at 37 ° C for 34 days) and in vivo (implanted into the mouse brain for 36 days) aged implants with new devices (Figure 2g–i). All characterization for aged devices was performed after the removal of spheroids. The number of working electrodes (impedance at 1kHz < 100 kΩ) remained unchanged for all aging groups (Figure 2g). We suspect that the reason the yield is not 100% in new devices already is mostly due to poor or broken electrical contact between the electrodes and the connector. Indeed, visual inspection under the microscope shows no broken tracks following the transfer fabrication process. A presumably simple fix would be to switch to a different type of connector to increase the yield. However, the impedance of the explanted electrodes has increased significantly (Figure 2h) when compared to new devices. In addition, the charge injection capacity (CIC) surprisingly increased significantly during in vitro aging but showed the expected decrease after in vivo implantation compared to new implants (Figure 2i). Taken together, these results suggests that although the expected biofouling happens during implantation the resulting electrical properties of the electrodes remain within the suitable range for the proposed stimulation. This indicates that the electrode-electrolyte interface is stable and that the electrodes do not dissolve or delaminate (a scotch tape test was performed after MEA fabrication and shows that the tracks are solidly bonded to the PDMS substrate, Figure S3, Supporting Information). We imaged a sample from each condition in a scanning electron microscope (SEM) to investigate the changes on the electrodes surface (Figure 2j). Some debris are visible on the surface of the electrodes aged in vitro. The origin of the debris is unknown, but we hypothesize that it is coming from the spheroids that were seeded on the electrodes. The post in vivo sample also shows adsorption of biological material on the surface of the electrodes. It is unclear whether this is coming from the spheroids in the device or the host animal.

2.3. Axon Growth Within the Implantable Biohybrid Neural Interface

The biohybrid concept requires that axons (1) grow toward the nerve channel and (2) avoid turning back or growing into neighboring seeding wells. We achieved this by introducing specific geometric features into the microstructure. In timelapse movies we tracked growth cones of growing axons using a convolutional

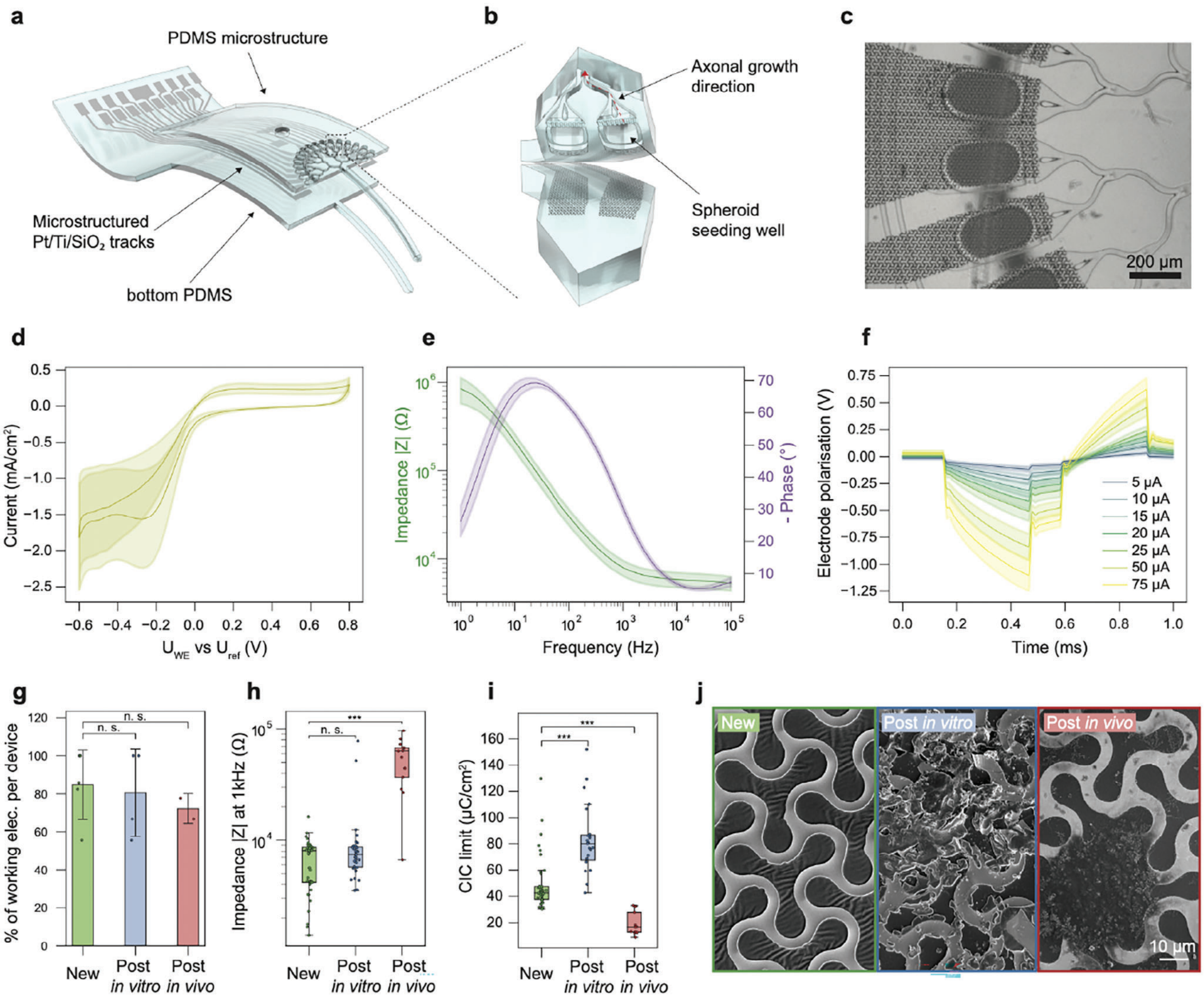

Figure 2. Microfabrication and electrochemical characterization of the biohybrid implant. a) Exploded view of the biohybrid device, comprised of three layers: a PDMS substrate, a multielectrode array, and a PDMS microstructure. b) Zoom-in on a seeding well area (dashed box) with the underlying microstructured electrodes. The axons of the spheroids seeded in the well area can grow along the microchannels (red arrow). c) Detailed micrograph of the electrodes, seeding wells and axonal guidance channels. d) Cyclic voltammetry (CV) of microelectrodes at 200 mV/s (n = 13 electrodes), exhibiting the electrochemical window between –0.6 and 0.8 V. e) Electrical impedance spectroscopy (EIS) of microelectrodes after microfabrication (before spheroids seeding), showing the module (green line) and phase (purple line) of the impedance versus frequency. n = 13 electrodes. f) Voltage response of microelectrodes to current-controlled biphasic pulses of 300 µs per phase with a 100 µs delay between phases. n = 13 electrodes, for every electrode and every current step 10 pulses were acquired and averaged. g) Comparison of the percentage of electrodes working in different conditions. New: devices after fabrication, In vitro: devices aged in vitro for 34 days. The devices were used once or twice for in vitro stimulation experiments. Ex vivo: devices measured when explanted after 36 days in vivo. ($P = 0.16$ and $P = 0.5$ for New vs in vitro and ex vivo respectively) h) Comparison of the electrode impedance at 1kHz in different conditions. ($P = 0.16$ and $P = 4.4\text{e-}7$ for New vs. in vitro and ex vivo respectively) i) Comparison of the charge injection capacity (CIC) of electrodes in different conditions. ($P = 1.5\text{e-}7$ and $P = 1.4\text{e-}7$ for New vs In vitro and ex vivo respectively) j) SEM images comparing the electrode surface at different aging stages. Left: sample after fabrication, middle: sample after aging in vitro, right: sample after aging in vivo. In d, e, and f data are mean (solid lines) ± standard deviation (shaded area). In h, data is mean (bars) and standard deviation (error bars). In boxplots, the median, quartile box and minimum and maximum values (excluding outliers) are shown. The significance was tested using a Mann–Whitney U test. The difference between two groups is considered significant if the P-value < 0.05, and *** denotes $P < 0.001$.

neural network and analysed how the critical parameters (e.g., channel width, number of loops redirecting axons, etc.) affect axonal growth speed and growth directionality (Figures S4 and S5), resulting in the axon guidance layout shown in **Figure 3a** and Figure S6 (Supporting Information). The axon guidance structure was entirely immersed in retinal ganglion cell (RGC) medium. The devices were plasma activated and subsequently coated with a 1:25 dilution of Matrigel in RGC medium, as described in the Experimental Section. Axon guidance is achieved through the edge guidance mechanism and no further chemoattractants were added. In a prior study,[37] we determined that the maximum achievable length of primary rat retinal axons within

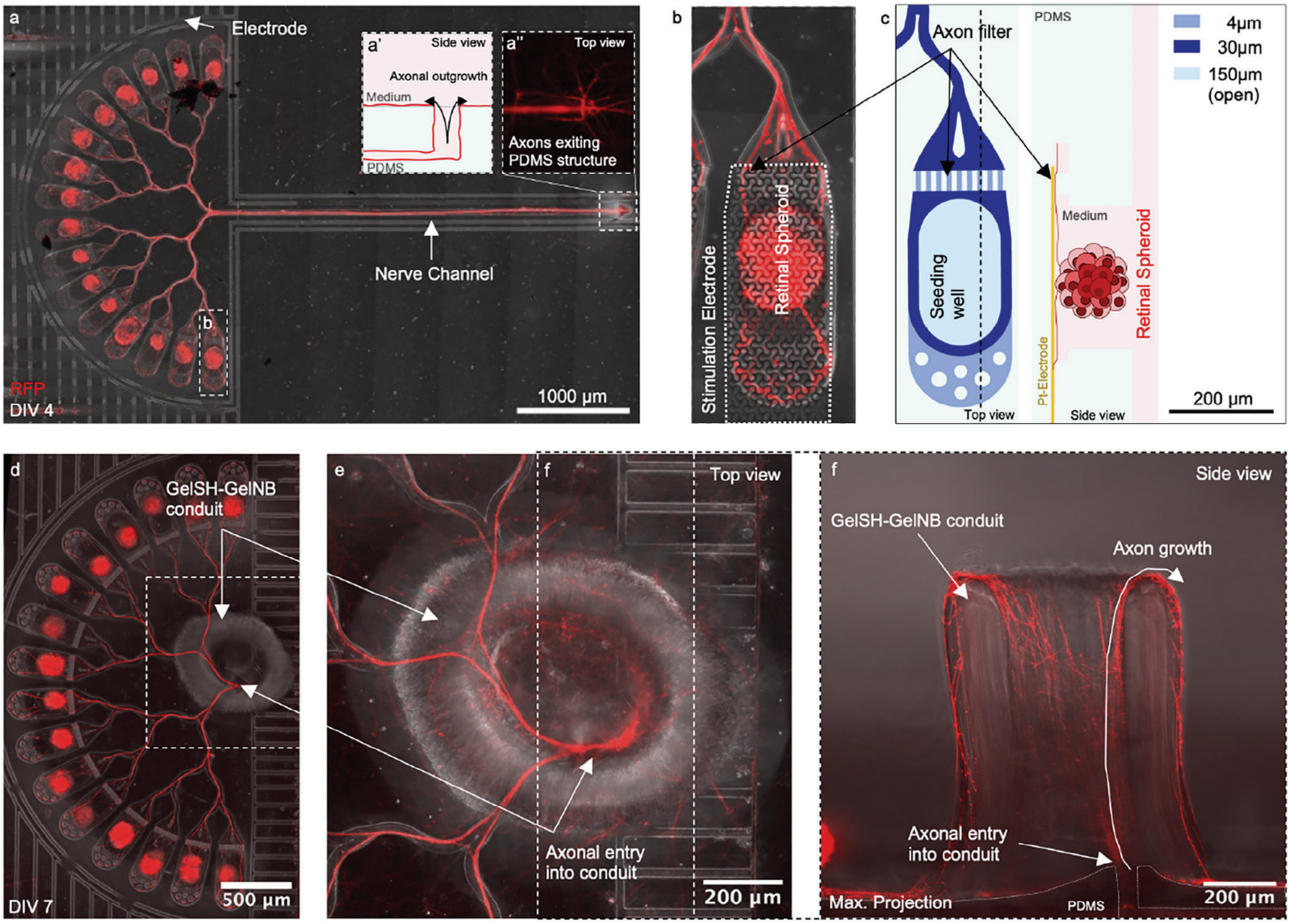

Figure 3. Axon growth within the biohybrid neural interface. a) Retinal spheroids extend axons through the matrigel-coated axon guidance structure, forming a nerve-like structure within 3.5 days after seeding. a′: Side view of the axon outlet at the end of the nerve channel. The triangular shape promotes axonal outgrowth while reducing axon re-entry into the channel. a″: Retinal axons exit the nerve channel at the axon outlet. b) One of the 16 seeding wells with an underlying electrode. Neural somata are blocked from entering the channel by an axon filter, which consists of 9, 4x10 µm small channels. c) Schematic showing top and side views of the seeding well. Retinal spheroids are manually seeded into each seeding well. d) A biohybrid implant where the PDMS channel was replaced by a GelSH-GelNB nerve conduit that is directly fabricated onto the axon outlet. e) Magnified top view of the hydrogel conduit. f) Side view maximum projection of the hydrogel conduit showing how axons exit the PDMS microstructure and grow upwards and along the inside of the conduit. Once they reach the top of the conduit, the axons continue to grow along the outside walls of the conduit since no target structure is present in vitro.

Laminin-coated PDMS microstructures was approximately 6 mm. Each of the 16 seeding wells contains 1-2 800-neuron spheroids consisting of primary embryonic rat retinal neurons. Neuronal spheroids extend axons through an axon filter consisting of 10 small channels with a cross-section of 10×4 µm (w × h) that prevent neural cell bodies from entering the channel system (Figure 3b,c; Figure S6c,d, Supporting Information). The axons entering the channel systems are directionally merged (Figure S6e–h and Movie S1, Supporting Information) toward a nerve channel. At the end of the nerve channel, a triangular axon outlet facilitates axonal outgrowth for potential target innervation (Figure 3a; Figure S6i,j, Supporting Information). Time-lapse recordings showed that axons enter the nerve channel in less than 16 h and reach the end of the 3.6 mm long nerve channel 44 h after seeding the retinal spheroids (Figure S7 and Movie S1, Supporting Information). Focused ion beam scanning electron microscopy (FIB-SEM) of the artificial nerve-like structure and subsequent segmentation showed that the axons form a bundle of up to 1200 axons including potential extracellular matrix fibers and side branches (Figure S8, Supporting Information).

Since in vivo the device requires the implantation of the nerve channel to guide the axons into the brain, the PDMS-based nerve channel will stay in the brain long-term. In order to improve the integration of the guided axons from the implant to the underlying tissue, a version of the device with a bioresorbable hydrogel[41,42] was developed, the rationale being that the hydrogel conduit would enable temporary guidance to the target region after which it gets absorbed by the tissue to enable integration and potentially also myelination of innervating axons. Using UV-curable gel-SH gel-NB we built hydrogel conduits of about 300 m inner diameter onto the axon outlet of the implant (Figure 3d; Figure S9, Supporting Information).[43] After 7 days, axons transited into the conduit and grew along the inside wall of the conduit (Figure 3e,f; Figure S10, Supporting Information).

Lateral force applied to the conduit shows that the bond to PDMS is strong enough to withstand the in vivo implantation procedure (Movie S3 and Figure S9, Supporting Information). Matrigel filled glass (Figure S11, Supporting Information) or collagen conduits (Figure S12 and Movie S2, Supporting Information) could also serve as alternative axon-guidance structures.

2.4. Spike Propagation Within the Biohybrid Microstructure

Electrically evoked action potentials need to reach their target and should not propagate into neighboring stimulating channels (crosstalk). In the presented implant, edge guidance-based mechanisms were used to direct axonal growth toward the nerve channel.[44] However, due to the inherent stochastic nature of axonal growth, relying solely on edge guidance introduces a degree of unpredictability, which sometimes results in axons growing into neighboring channels instead. If axons are misguided and grow into neighboring channels it can lead to crosstalk between individual seeding wells. Crosstalk means that neurons of one seeding well could synaptically induce activity in the neurons of the neighboring channels (not to be confused with crosstalk induced by the electrode of one seeding well also directly stimulating the neurons of another neighboring seeding well). This unintended crosstalk might reduce the stimulation resolution of a biohybrid implant and should be avoided. We used high-density CMOS-based MEAs (HD-MEAs) as a tool to assess how stimulation induced spikes propagate within the geometry of the biohybrid microstructure. For these experiments, the microstructure with the nerve-forming channel was laid flat onto the sensing area of the HD-MEA (**Figure 4a**), coated with 1:25 matrigel and completely submersed in 1 ml RGC medium. The voltage map[45] confirmed proper adhesion onto the CMOS electrodes (Figure S13a, Supporting Information) and confined axonal growth within the axon guidance structure for up to 167 days (Figure 4b). Based on timelapse recordings axons showed substantial growth during the initial 4 days in vitro. In all tested samples where coating was effective, axons reliably extended to the end of the nerve channel. However, individual axonal variation was not quantified, and the long-term effects on axonal integrity were not adressed in this study. We assessed the propagation of electrically evoked spikes within the biohybrid structure ($n = 1$) by repeatedly stimulating (250×) the axons at each of the 16 source channels and recording the corresponding elicited action potential events within the following 3.5 ms at both the source channels and at defined locations along the nerve channel. The stimulation response matrix can be seen in Figure 4c. The location of the recorded spikes in Figure 4d,e,f was color coded according to Figure 4a. We can distinguish three different signal propagation trends: 1) Stimulation-induced spikes propagate from the stimulated source channel toward the end of the nerve channel without traveling toward the end of a neighboring source channel (no crosstalk) (Figure 4d, d'; Movie S4, Supporting Information). (2) Spikes reach the target channel but also enter into a neighboring source channel (Figure 4e, e'; Movie S5, Supporting Information) or (3) only propagate into another source channel without reaching the target channel thereby causing crosstalk between the source channels (Figure S13d, Supporting Information). Although individual channel stimulation should be sufficient to assess spike propagation into the nerve channel we wanted to confirm results by stimulating the nerve at the location "Target 3" (Figure 4a) and observed that spikes traveled into 13 out of 16 source channels (Figure 4f, f'; Movie S6, Supporting Information). We added three stimulation targets to evaluate how far spikes propagate within the nerve channel. In summary, 12 out of 16 stimulated channels could elicit a response in the nerve channel (in at least one out of the three target locations) at DIV110, with 7 out of those 12 also eliciting unwanted crosstalk. 4 out of 16 channels showed measurable crosstalk (Figure 4c) and no transmission to the nerve channel. The fluorescence image reveals that axons from channels 11, 15, and 16 did not reach the nerve channel, which correlates with the absence of spike propagation into the nerve channel for these cases. Stimulation of two biological replicates at DIV 27 showed a target response including crosstalk from a lower number of source channels (1/16 and 6/16) (Figure S13c, c'; Supporting Information).

2.5. In Vitro Stimulation of the Implantable Neural Interface

After demonstrating that stimulation-induced spikes can propagate toward the nerve channel using HD-MEAs, the aim of this experiment was to assess whether the stretchable microelectrode array in the biohybrid implant could successfully stimulate the neural spheroids within the seeding wells. A PDMS well for culture media (200 µL) was glued around the microstructure to prevent medium from entering the omnetics connector (**Figure 5a**). Devices were coated with 1:25 matrigel and subsequently submersed in RGC medium. To prevent evaporation all PDMS wells were covered with a glass coverslip. The spheroids grown in the device can be seen in Figure 5b. Neural activity of spheroids in the implant was measured by functional Calcium imaging. Electrical stimulation of working electrodes (manually timed trigger, 2 V peak to peak, 200 Hz, parameters based on prior in vitro experiments[37]) resulted in a consistent calcium response (Figure 5c,d; Figure S15f,g–j, Supporting Information). Electrode size in the different implant versions varied and either stimulated a single seeding well (Figure 5b; Figure S15e, Supporting Information) per electrode or up to two seeding wells per electrode (Figure 5b; Figure S15h, Supporting Information). Stimulated GCaMP fluorescence propagated along the main branch (blue traces 2-4) but did not show any detectable signal in the side branches (red traces 5-6) or the neighboring seeding wells (red traces 7-8) (Figure 5c) which means that we did not measure any crosstalk. We did not systematically stimulate and measure the Calcium response of the other seeding wells as we had already done systematic crosstalk analysis on the CMOS array. However, neural activity could be induced in multiple electrodes (Figure S15, Supporting Information), which is important for stimulating any target structure with different stimulation patterns. In vitro tests demonstrated that the stretchable MEAs effectively stimulate neural spheroids and that the biohybrid structure can direct electrically evoked spikes toward the nerve channel. Although structural guidance minimized crosstalk it is alone insufficient to prevent it completely. Stimulation parameters were chosen such that the fluorescent response was large enough to be recorded by our system. In the future, a careful investigation of voltage-controlled stimulation parameters is needed to ensure

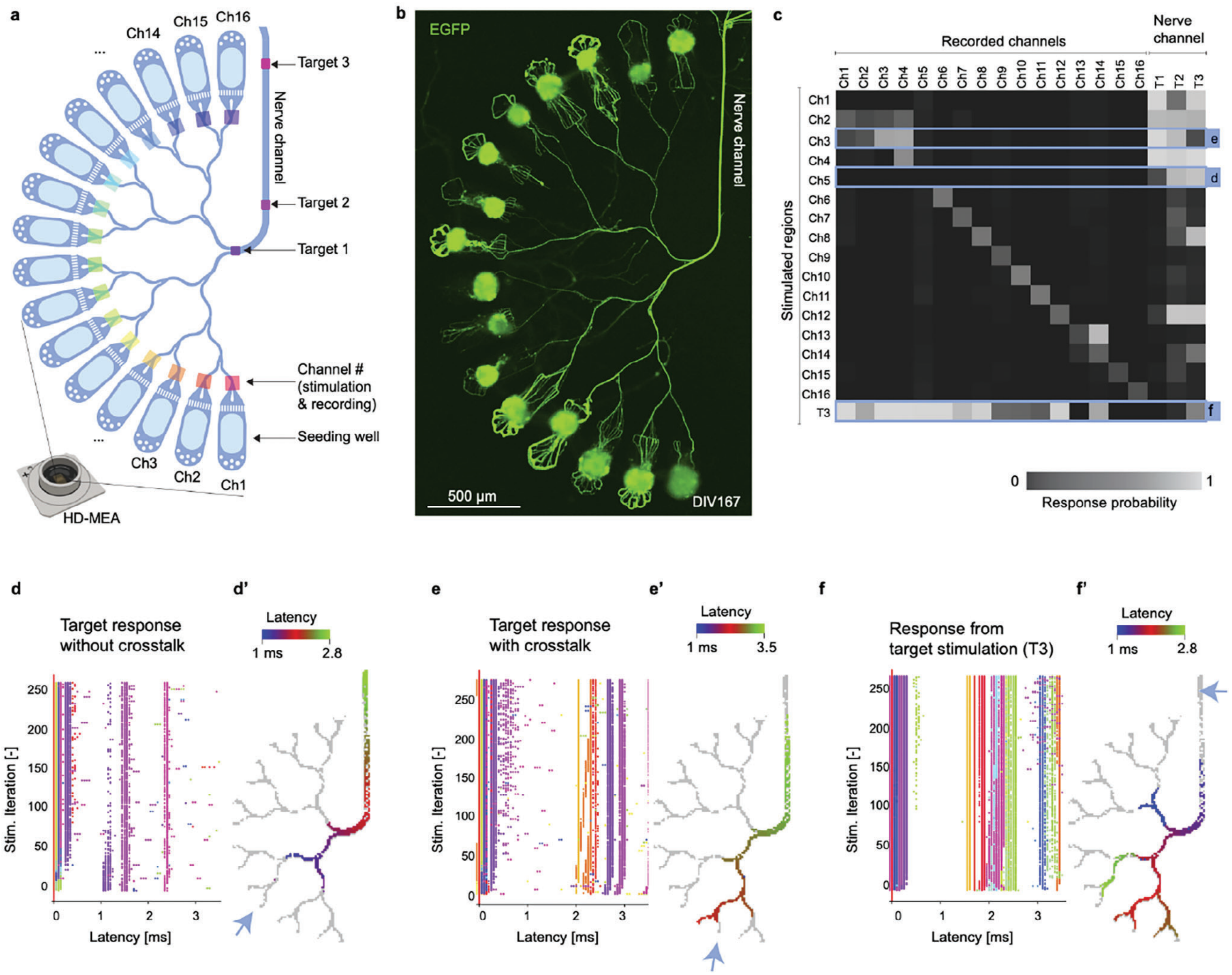

Figure 4. Measuring electrically evoked action potentials within the biohybrid axon guidance structure to assess target response. a) Schematic of the biohybrid axon guidance structure mounted onto the HD-MEA. Colored rectangles indicate the location of specific stimulation and recording channels and regions. b) Fluorescence image showing axonal growth of retinal spheroids within axon guidance structure mounted on the HD-MEA at DIV167. c) Stimulation response matrix shows different response types (target response and crosstalk) within the biohybrid structure at DIV110 (n = 1). Response probability is the normalized number of detected spikes in each recorded channel or region after stimulation. d–f) Stimulation induced raster plot (SIRP) of spikes propagating into the nerve channel. Colormap indicates channel location according to subpanel a. d) response without cross talk. e) response with cross talk. f) Retrograde response upon target stimulation. (d′, e′, f′): Latency map outlining the elicited response propagation within 3.5ms. Blue arrow indicates stimulated channel. Grey pixels indicate all recorded electrodes. (d′): response without cross talk. (e′): response with cross talk. (f′): Retrograde response upon target stimulation. Presented data is from n = 1 biohybrid axon guidance structure. Analysis of 2 more cultures can be found in Figure S13 (Supporting Information).

that we are operating within the water window and that no water splitting occurs, which might also elicit neural activity.

2.6. In vivo Implantation of the Biohybrid Implant

The feasibility of the implantable biohybrid neural interface was tested in a living mammalian model by implantation onto the surface of the cortex and targeting the visual thalamus. This involved overcoming numerous challenges due to the difficulty of fully controlling all relevant parameters within an in vivo environment, such as nutrient flow, oxygen level, temperature, and sterility. After performing a small craniotomy over the left primary visual cortex in 30–60 day old B57BL/6 mice, the implant was inverted in a way that the neural spheroids were facing the brain (**Figure 6**a,b). In order to minimize the risk of excessive bleeding, brain damage and stronger immune response, all except one device were implanted epidurally (Table S1, Supporting Information). Additionally, we reasoned that following an epidural implantation the dura would provide a physical barrier between neurons from the brain and neurons from the device, thereby reducing the risk of direct connections forming between them. We prevented neural spheroids from falling out of the implant during implantation by submersing the implant in a high viscosity

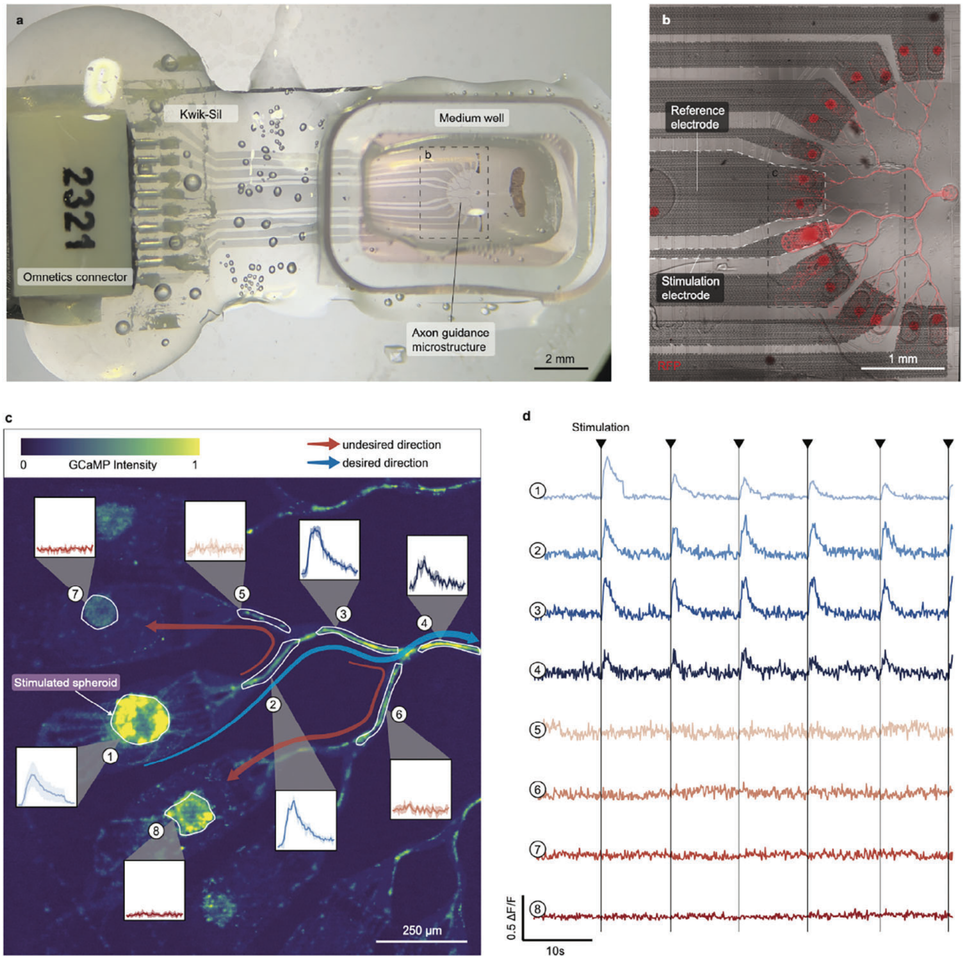

Figure 5. Electrical stimulation of neural spheroids in the biohybrid implant. a) Biohybrid implant with a medium well mounted on top of the axon guidance structure for in vitro stimulation of seeded neurons during functional calcium imaging. b) Axon guidance structure with underlying microelectrode array at DIV11. The multi-electrode array has 8 stimulation electrodes and 1 reference electrode. RFP labelled spheroids were grown within the microstructure. Image was log processed to enhance axon visibility. c) The neural spheroids were labeled with the calcium indicator GCaMP8m. One spheroid was repeatedly stimulated (2 V peak to peak, 200 Hz) and the corresponding calcium response inside the spheroid and along the channels was measured. Blue arrow indicates desired signal propagation. Red arrow indicates undesired signal propagation. d) Repeated electrical stimulation (arrows) induced a calcium response in the neural spheroid (trace 1) and along the axonal main branch (traces 2,3,4). No calcium response could be detected in neighboring spheroids (traces 7,8) or side branches (traces 5,6). Blue traces indicate traces recorded along the desired signal propagation path and red traces along undesired paths. Data from n = 1 culture.

medium containing 1% methylcellulose. While lowering the implant manually onto the cortical surface, the PDMS nerve channel was inserted through a physical path, which we preformed using a 0.45 mm diameter injection needle reaching 2.8 mm deep, toward the dorsolateral geniculate nucleus (dLGN) in the visual thalamus (Figure S16a, Supporting Information). Therefore, only the nerve channel was inserted deep into the brain, while the main implant containing the spheroids remained in place on top of the cortical surface. A 4-mm glass window together with a headbar were fixed on the cranium to protect the device and enable head fixation under the two-photon microscope (Figure 6a–c). Mice integrated the implant successfully with a 100% survival rate post surgery and showed no behavioral impairments in the days and weeks following the surgery (Movie S9, Supporting Information). No immunosuppressive medications were given to the animals during the in vivo experiments, besides the nonsteroidal anti-inflammatory drug Carprofen, which was given for 48h following the implantation surgery.

Next, we investigated how epidural implantation affects viability and axonal growth of implanted spheroids. The biohybrid device was implanted either within 24 h after seeding (DIV0/DIV1) or after the in vitro formation of the nerve-like fascicle at DIV7. Implanting a predeveloped nerve was intended to facilitate quicker axonal integration into the host tissue and to allow for the selection of the most robustly growing implants prior to implantation. In vivo two-photon fluorescence imaging (Figure 6c) showed that in DIV0 or DIV1 seeded implants, spheroids extended axons for 4–5 days after implantation in 8 out of 9 imaged mice, with axons reaching the nerve channel in 6 cases (Figure 6d-g; Figure S16b and Table S1, Supporting Information). The axonal growth rate was very similar to what we observed in vitro in which axons formed an in vitro nerve-like

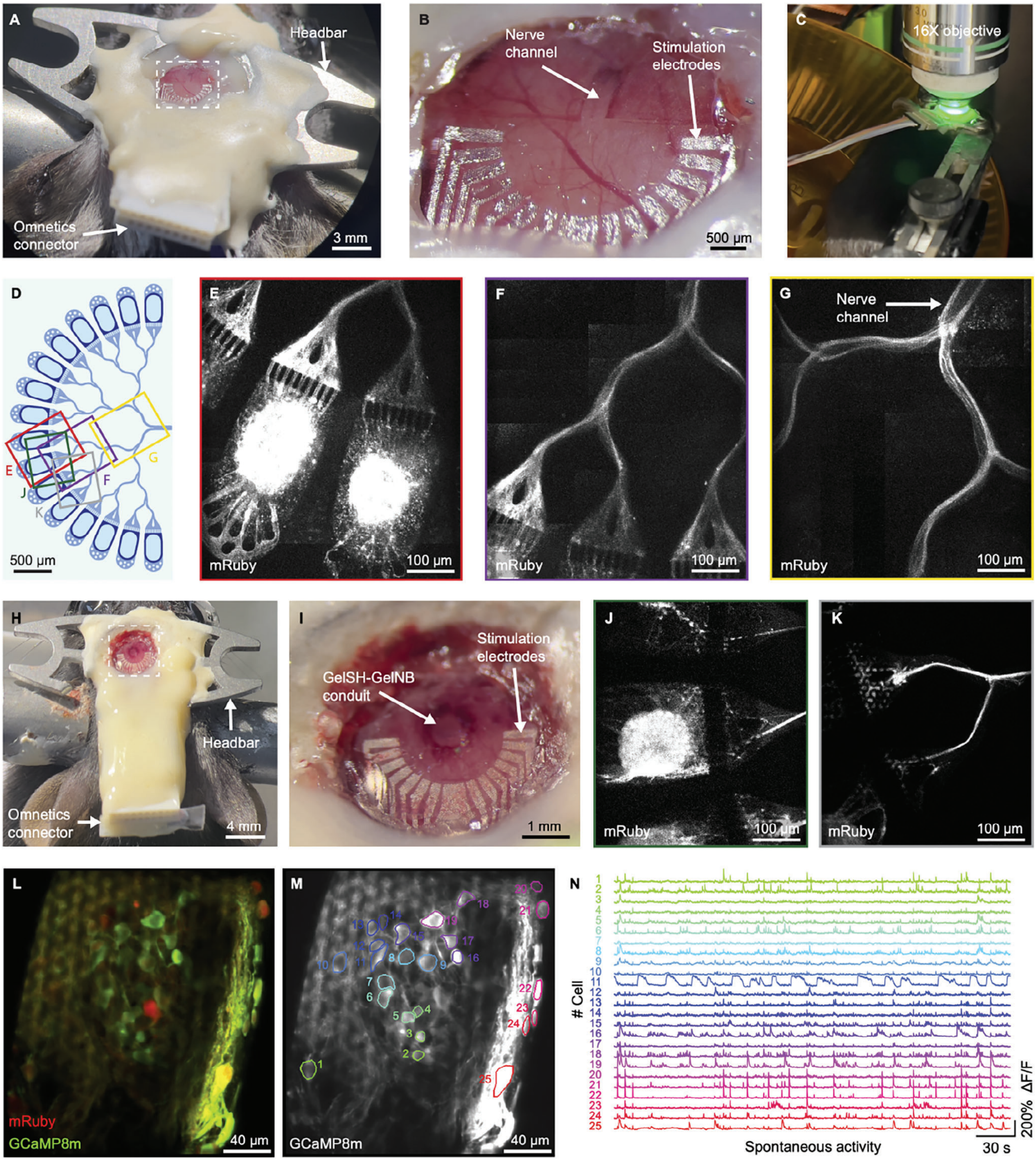

Figure 6. Implantation of the biohybrid implant. a) Mouse with biohybrid implant and PDMS nerve channel. b) Zoom from *a* showing the biohybrid implant on the cortex. The nerve channel enters the brain at the level of the dLGN. c) Awake two-photon imaging setup. d) Schematic layout of the implanted axon guidance structure. e) Two-photon image of cortical spheroids in the seeding wells four days after implantation. f) Two-photon image of axonal growth 4 days after implantation. g) Two-photon image of axonal growth into the nerve channel 4 days after implantation. h) Biohybrid implant with the GelSH-GelNB conduit. i) Zoomed from *h* showing the biohybrid implant fixed on the top of the exposed cortical surface. The GelSH-GelNB conduit enters the brain at the level of the dLGN. j) Cortical spheroids in the seeding wells of the GelSH-GelNB conduit 4 days after implantation. k: Two-photon image of axonal growth. l) Cortical spheroids expressing mRuby and GCaMP8m 22 days after implantation. m: GCaMP8m expression with ROIs for quantifications shown in n. n) Spontaneous calcium activity traces of individual neural spheroids in vivo at 22 days after implantation. Calcium activity serves as a proxy for neuronal activity. Neurons display complex spontaneous activity patterns with different spheroids having distinct calcium dynamics, suggesting that they form a healthy, interconnected neural tissue.

fascicle within 3–4 days. The spheroids showed intact fluorescent neural morphologies for up to 22 days after implantation (Figure S16c and Table S1, Supporting Information). Devices implanted with a pre-developed 7 days old nerve structure showed intact neural structures when imaged directly after implantation (Figure S16d–f, Supporting Information). For both implant strategies, axonal growth and intact axonal morphology were observed for up to five days (Table S1 and Figure S16h,i, Supporting Information). Beyond this period, an autofluorescent (green channel) fibrous structure emerged replacing the red fluorescing axons (Figure S17a, Supporting Information). Post-hoc immunochemistry showed the presence of astrocytes and microglia in both the seeding wells and nerve structure (Figure S17b–d, Supporting Information), indicating that neuroinflammation could be the cause of axonal degeneration and the source of the fibrous structure observed in vivo.

After demonstrating the successful implantation of devices with PDMS nerve channels, we could then show that implants with a GelSH-GelNB conduit instead of the PDMS nerve channel were also sufficiently robust for in vivo implantation (Figure 6h,i; Figure S18a–c, Supporting Information). Four days after implanting these devices with a similar strategy as described before, neural spheroids showed strong mRuby expression (indicative of living neurons), Figure 6k; Table S1, Supporting Information) and axonal growth within the axon guidance structure (Figure 6j; Table S1, Supporting Information). Post-mortem histology showed that the GelSH-GelNB conduit was still present in the brain after 22 days (Figure S18a–c, Supporting Information) of implantation.[41,42]

Finally, we recorded calcium (GCaMP8m) activity, which serves as a proxy for neuronal activity, of the neural spheroids in vivo (Figure 6l). We found complex spontaneous activity patterns across spheroids, with different spheroids having distinct calcium dynamics, suggesting that they form a healthy, interconnected neural tissue. (Figure 6m,n). These recordings demonstrated the viability and neuronal function of the spheroids for up to 22 days. Given that using our biohybrid implant we have not yet achieved axonal integration into the host brain, the observed calcium activity is very likely spontaneous and not indicative of neural communication between the implanted spheroids and the host tissue.

Overall in vivo implantations demonstrate that neurons show initial healthy axonal growth down into the nerve channel for up to 5 days and good viability within epidurally implanted biohybrid devices for up to 22 days.

3. Discussion

In this work, we developed an implantable biohybrid neural interface toward synaptic deep brain stimulation. The concept is centered on electrically and functionally isolating individual neural spheroids within an implantable nerve structure in order to exploit neurons as synaptic relays. We believe that future developments of the biohybrid strategy have the potential to increase the stimulation resolution of currently available deep brain stimulation electrodes because of the potential for targeted synaptic stimulation and improved long term biocompatibility as only axons will integrate and interact with the host brain.

The biohybrid device was fabricated following a straightforward process flow. Once the template is fabricated, it can be reused several times to perform the template-stripping transfer-printing procedure, which leads to straightforward fabrication of MEAs on PDMS. The devices showed satisfying performance, highlighted by good electrochemical properties. The yield of working electrodes per device should be improved, but most of the non-working channels can be traced back to the interface between the omnetics connector and the Pt tracks. Future iterations might benefit from using a flexible cable connector such as that used in Fallegger et al.[46] to increase the number of working electrodes. When performing the aging experiment, the impedance of the explanted electrodes after being implanted in vivo increases, as expected and reported in different works.[30,47,48] However, the impedance increase saturates at an acceptable level after 4 weeks in vitro with periodic stimulation. No change in impedance was noted for the post in vitro samples, despite the presence of debris seen on SEM pictures. A potential reason for this behavior could be delamination of the electrode passivation, or intrusion of water under the passivation layer. Further tests are needed to fully elucidate the phenomenon. The CIC of the electrodes also exhibits values similar to what can be found in the literature for flat Pt[49,50] along with the expected decrease after in vivo aging due to physical processes such as biomolecule absorption, fundamental differences in counter-ion transport, and diffusional limitations on charge transfer.[47,51] The stimulation pulses used during in vitro and in vivo experiments, which were voltage-controlled with a 2 V amplitude, might also explain partly the change in impedance and CIC after aging. Since voltage-controlled stimulation lacks precise control over the electrode-electrolyte interface voltage, operation may have exceeded the water window boundaries. Additional studies are needed to establish voltage thresholds that preserve long-term neuronal viability within the device. Alternatively, switching to a current-controlled setup would solve this issue as well.

Achieving highly reliable axonal guidance from the source seeding to the nerve channel is essential for a high-resolution biohybrid neural interface. Any misdirected axon will result in the loss of an electrode brain connection and might cause potential synaptic crosstalk with other channels within the biohybrid implant. We used edge guidance-based mechanisms to direct the axons into the desired direction.[44] The inherent stochastic characteristics of axonal growth introduce a level of unpredictability in directing axonal growth when only relying on-edge guidance mechanisms.[52] This may lead to signal propagation into neighboring channels and potential synaptic crosstalk. Indeed our stimulation experiments on CMOS MEAs confirmed that spikes not only propagate into the nerve channel but also into neighboring seeding wells. It remains to be tested if the observed crosstalk can elicit secondary spikes in neighboring channels that would negatively affect the channel selectivity and stimulation resolution. Moreover, future studies need to analyze if and how confining non-myelinated axons into an artificial nerve like structure leads to any synaptic or non-synaptic crosstalk between the axons themselves. Introducing myelination through co-seeded oligodendrocytes could potentially resolve potential crosstalk between axons and result in faster axonal spike propagation. However, the increased metabolic activity introduced by the added oligodendrocytes could lead to nutrient or oxygen shortages within the microchannels and needs to be assessed for feasibility. The low number of seeding wells and the high number of neurons per seeding well in the here presented model required only four to converge axons from all seeding wells into the nerve channel. Therefore, the guidance motifs used in our model were sufficient to form a nerve-like structure. For future versions of the biohybrid interface, however, one should reduce the number of neurons per seeding well (ideally 1 per electrode) and increase the number of electrodes. This will provide non-redundant high-resolution sensory input to the brain in which a single electrode can modulate the activity of individual neurons. Further research will be needed to develop suitable methods that enable culturing of isolated single neurons without compromising their viability. Depending on the merging architecture, this will require each axon to cross a significantly higher number of channel merges, which will increase the chance of axons turning into the wrong channel. Thus, future biohybrid neural interfaces will have to incorporate additional axon guidance approaches that are, e.g., based on molecular gradients used to direct axons in vivo[53,54] or electric cues.[55] To summarize, relying solely on edge guidance based mechanisms to guide axons as presented in this study will not be sufficient to achieve the intended almost single neuron resolution for synaptic brain stimulation. The CMOS arrays presented here will be a suitable platform for functionally testing new guidance strategies in future studies.

The introduction of hydrogel-based conduits into the implant was intended to enhance biocompatibility over a PDMS based nerve channel and enable integration of axons within the neural tissue. PDMS, being a synthetic material with higher mechanical stiffness compared to biological tissues will likely lead to inflammation over time when permanently implanted and thereby compromise the functionality of the neural interface. To overcome these disadvantages, we incorporated hydrogel conduits for temporary axon guidance. The term “temporary guidance” implies that the hydrogel conduit provides an initial pathway for the axons but is not intended to remain permanently. Over time, the hydrogel is expected to degrade and be absorbed by the body.[56,57] The temporary nature of the hydrogel conduit enables oligodendrocytes to access axons and potentially myelinate them over time. Moreover, it enables vascularization of the newly formed nerve over time which could enhance viability of the biohybrid nerve. Both PDMS and hydrogel-based conduits can be fabricated several centimeters long, sufficient for targeting structures like the visual thalamus in humans, which lies approximately 5 cm below the cortical surface. However, the stiffness of these conduits, when extended to such lengths, is likely insufficient for unaided insertion. For rodent implantations, conduits up to 3 mm in length were sufficiently rigid for insertion into the brain along a preformed path created using an injection needle. In contrast, for deep brain implantations in humans, a temporary stiff guidance structure—either inserted into the conduit lumen or encapsulating the conduit—would be required to guide the conduits to their target structures.[58]

In vivo we have found that neural spheroids were able to grow and extend axons down into the nerve channel within the first 3 days of epidural implantation. However, about 4 days after implantation, axons started to degrade with a strong increase in autofluorescence along the axon guidance paths and seeding wells. This autofluorescence may be due to degrading neuronal structures or a triggered immune response with, e.g., phagocyting macrophages or microglia[59] around the developed neuronal structures,[60] which would be consistent with the observed astrocytic and glial markers in the explanted devices. As we implanted rat neurons into wild type mouse brains, neuronal death could be the result of an immune response or because of an increased demand for oxygen and nutrients from developing neurons that cannot be matched by the passive diffusion of nutrients through the meninges. Neuronal long-term viability might benefit from subdural implantation as neurons would get a better access to neurotrophic factors released form the brain. Future studies might benefit from subdural implantation into immunodeficient mice or rats, as direct transplantation of neurons, spheroids, or organoids into such models has previously led to successful neuron survival and neural network integration in vivo.[33,35,36] In future studies, we intend to use immunodeficient rodents and/or immunosuppressive medications to avoid immune responses in preclinical in vivo experiments. For future human applications, the biohybrid technology will require the use of patient iPSC derived neurons to prevent immune rejection by the host tissue.

4. Conclusion

In summary, we have presented and characterized an implantable neural interface showing first feasibility toward in vivo application. Although we have yet to demonstrate axons transitioning from the implant into neural tissue to form functional synapses in vivo (our previous work has shown some initial feasibility of this approach in vitro[37]), we present promising initial findings on neural viability and axonal growth after implantation. Future investigations will need to determine how the in vitro feasibility observed translates to in vivo conditions, with particular focus on axonal growth dynamics within adult CNS tissue and the formation of functional synapses. In future clinical applications of the biohybrid technology patient specific iPSC derived neurons can be integrated into the biohybrid device to provide personalized sensory restoration.

5. Experimental Section

Primary Neuron Cultures—Primary cell source and cell dissociation: Primary cortical and retinal cells from E18 embryos of pregnant Sprague-Dawley rats (EPIC, ETH Phenomics Center, Switzerland) were used for all experiments in compliance with 3 R regulations. Previous to the experiments, approval was obtained from Cantonal Veterinary Office Zurich, Switzerland, under license SR 31175 - ZH048/19. Briefly, E18 time-mated pregnant rats (Janvier Laboratories, France) were sacrificed, and the embryos were removed. Embryonic eyes and cortex tissue were dissected and stored in Hibernate medium on ice.

Tissues were enzymatically digested using a Papain solution (50 mg Bovine serum albumin (BSA) (A7906, Sigma-Aldrich) and 90.08 mg D-glucose (Y0001745, Sigma-Aldrich) in 50 mL sterile PBS, and vortexing 5 mL of the solution with 2.5 mg Papain (P5306, Sigma-Aldrich) and 5 l DNAse (D5025, Sigma-Aldrich)).

The Hibernate medium was carefully aspirated from the tubes containing retinal and cortical tissue. 2.5 mL of the prepared Papain solution were added into each tube. The tubes were incubated for 15 min at 37 °C and gently shaken every 5 min. The Papain solution was aspirated without disturbing the pellets, and 5 ml warmed Neurobasal medium (21103049, Gibco, Thermo Fisher Scientific) with 5 % B27 supplement (17504044, Gibco, Thermo Fisher Scientific) and 10 % Fetal bovine serum (FBS) (10500056, Gibco, Thermo Fisher Scientific) was added. After 3 min of incubation, the media was removed. The incubation and aspiration steps were repeated twice using Neurobasal medium with 5 % B27 supplement. After the last aspiration, 2–4 mL warmed RGC medium (medium as in Ref. [37]) were added to the retina and to the cortex tissue tube. Lastly, tissues were mechanically dissociated using a 1 mL pipette until complete dissociation of the tissue.

Spheroid Generation: Commercially available AggreWell 400 microwell plates (AggreWell 400 24-well plate, 34415, StemCell Technologies) were coated with 500 μL of AggreWell Anti-adherence rinsing solution (7010, StemCell Technologies) and rinsed with 2 mL Neurobasal medium. The medium was replaced with 1 mL RGC (for primary embryonic E18.5 rat retinal spheroids) or Neurobasal medium (for primary embryonic E18.5 rat cortical spheroids) and pH-equilibrated in the incubator before seeding 960 000 cells per well to achieve a spheroid size of 800 cells per spheroid. Immediately after seeding the cells were transfected with an adeno-associated virus (AAV). Retinal and cortical spheroids were transfected with a mRuby, EGFP or GCaMP8m expressing AAV (scAAV- DJ/2-hSyn1-chI-mRuby3-SV40p(A), ssAAV-DJ/2-hSyn1-jGCaMP8m-WPRE-SV40p(A), ssAAV-retro/2-hSyn1-chI-EBFP2-WPRE-SV40p(A). All adeno-associated viral vectors were provided by the Viral Vector Facility of the University of Zurich. For producing homogeneously sized spheroids, the AggreWell plate was centrifuged at 100 g for 3 min and transferred into the incubator at 37 °C with 5 % CO_2. Spheroids were ready for seeding after 24 h.

Biohybrid Implant Fabrication–Fabrication of the PDMS multielectrode array: The entire process flow is illustrated in Supplementary Fig. S1. The MEA tracks were first fabricated on a Si wafer and subsequently transferred on PDMS. First, a 525 μ m thick 4 in Si wafer (Microchemicals GmbH) was patterned using standard photolithography (photoresist AZ ECI3102, spincoating 3 krpm for 30 s, softbake 60 s at 90 °C; exposure in EVG 620NT (EV Group), i-line exposure 110 mJ cm^{-2}, post-exposure bake 60 s at 11 °C; development 60 s in AZ726 MIF, hardbake 120 s at 110 °C) and RIE etched (Oxford NGP80 (Oxford Instruments, 10 min at 100 W, 100 mTorr, 30/12/10 sccm SF6/O2/CHF3) to create the microstructured tracks layout. The wafer was then cleaned in DMSO at 80 °C for 30 min, rinsed in DI water, and briefly plasma-cleaned (3 min at 250 W, 1 mbar, Technics Plasma 100-E (Technics Plasma GmbH)). A second photolithography and etching step was carried out to delimit the individual devices (photoresist AZ P 4620, spincoating 3 krpm for 40 s, softbake 180 s at 110 °C; broadband exposure 500 mJ cm^{-2}; development 4 min 30 s in AZ2026 MIF, hardbake 360 s at 110 °C; etching in PlasmaPro Estrelas 100 (Oxford Instruments), Bosch Process to etching depth ∼ 100 μm). The wafer was then cleaned in a similar fashion as before. Prior to the material stack evaporation, the wafer was silanized (Trichloro(1H,1H,2H,2H-perfluorooctyl)silane, Sigma-Aldrich 448931), vacuum-assisted deposition) to create an anti-adhesive layer that is crucial for the subsequent transfer process. Then, a stack of SiO_2/Ti/Pt (25nm/5nm/100nm) was evaporated on the wafer (E-beam evaporation, Plassys MEB550S). The MEAs were then stripped from the SI wafer and were transferred on PDMS following a procedure described in Ref. [38]. Briefly, a 15 % solution of PVA (Mowiol 18-88, Sigma-Aldrich 81365) in water was spin-coated onto a 125 μm thick PEN foil (Coloprint tech-films GmbH) and cured at 60 °C for 10 min. The patterned Si wafer with the SiO_2/Ti/Pt stack was heated up to 120 °C on a hotplate and the PEN/PVA was laminated onto it using a silicone roller while applying minimal pressure. At room temperature, the PEN/PVA was stripped from the wafer, effectively picking up only the patterned MEAs. The receiving substrate for the transfer was prepared by silanizing a 4 in wafer, followed by spin-coating PDMS (Sylgard 184, 1:10) at 500 rpm for 30 s and curing it in an oven at 80 °C overnight. Once cured, the PDMS wafer and the PEN/PVA with the patterned SiO_2/Ti/Pt were plasma activated (Tergeo Plasma (PIE Scientific), 35 W, duty ratio 50/255, 5 sccm O_2 10 sccm H_2O, 25 s) and brought into contact to be bound together. The assembly was immediately heated up to 120 °C on a hotplate (and weighted down to ensure good contact for bonding) for at least 30 min. At room temperature, the PEN foil was peeled off the PVA. The PDMS wafer with PVA was then immersed in boiled water for 2 times 5 min with a DI water rinse in between to dissolve the PVA. The MEAs on PDMS were then cut manually using a scalpel and were carefully lifted off the support wafer for subsequent handling.

Design of the PDMS Microstructures: PDMS microstructure layouts were designed using AutoCAD (Autodesk). The layouts were sent to WunderliChips GmbH (Zurich, Switzerland) for the fabrication of a master template wafer, as well as PDMS replica that were extracted to remove uncured monomers using ethyl acetate.

Assembly of the Biohybrid Implant for In Vivo Implantation: First, the PDMS MEA and microstructure were assembled using plasma bonding. Both were subjected to a plasma treatment (Tergeo, 35 W, duty ratio 50/255, 5 sccm O_2 10 sccm H_2O, 40 s), then a drop of water (∼20μL) was dispensed on the MEA and the microstructure was manually aligned on top using a stereomicroscope. The assembly was heated at 100 °C on a hotplate for ∼ 30 min to dry the water and ensure a good bond. An omnetics connector was aligned by hand and glued on the pads of the MEA using a silver conductive paste (Sigma-Aldrich 901769) and cured at 100 °C for 1 h. The connection points were then encapsulated in a soft silicone (Kwik-Sil, World Precision Instruments). Lastly, the device was cut to shape using a scalpel. Immediately prior coating and seeding, the whole device was plasma activated.

Assembly of the Biohybrid Implant for In Vitro Stimulations: The devices for in vitro testing were fabricated in a similar fashion as the devices for in vivo implantation up to the connector gluing and encapsulating. Then, the device was glued on a glass slide and a PDMS frame was glued around the microstructure to act as a reservoir for culture medium (using Kwik–Sil as glue). Immediately prior coating and seeding, the whole device was plasma activated to increase hydrophilicity and thereby ensure that the coating solution and culture medium enter the axon guidance channel system.

Coating and Spheroid Seeding for In Vivo Implantation: The implants were coated with 1:25 matrigel dissolved in RGC medium. A drop of matrigel was applied only at the nerve outlet of the microstructure to minimize surface coating and capillary forces enabled complete filling of the microstructure. The implant was then immediately submerged in RGC medium and equilibrated in a 37 °C 5 % V/V CO_2 incubator.

Flight Based gel-NB gel-SH Conduit Fabrication: We used thiol-norbornene clickable resins based on our previous work for the fabrication of the conduits.[61] A thiol-ene click chemistry-based system is less prone to network defects and swelling and the crosslinking chemistry is faster, when compared to chain-growth polymerization-based hydrogels such as gelatin methacryloyl (GelMA). In the present work, we used 2.5 % w/v thiolated gelatin (GelSH) and 2.5 % norbornene-functionalized gelatin (GelNB) in phosphate buffered saline (PBS). Lithium phenyl (2,4,6-trimethylbenzoyl) phosphinate (LAP) at 0.05 % w/v was added as the photoinitiator. We used the prototype FLight device developed in-house at ETH Zurich. The device is based on the filamented light fabrication technique,[43] and relies on optical self-focusing and modulation instability to fabricate cm-scale constructs featuring highly aligned microfilaments. These microfilaments are prevalent throughout the length of the constructs and are excellent guidance cues for cell alignment and proliferation. For the fabrication of the devices, the PDMS devices were first plasma activated (Tergeo plasma cleaner, Oxygen, 1 min 10 s, 25 Watt), were placed in 2-well ibidi dishes (ibidi, 80286) and positioned in the Flight projection system capable of a top-down projection of filamented light. The projection images (concentric circles equaling the cross-section of the conduits) were positioned such that the open lumen of the conduits was perfectly aligned with the open channel of the PDMS devices. Next, the devices were submersed in the resin (depending on conduit length 1-2 mm), followed by a 8 s long light exposure at 64 mW cm−2 to fabricate the conduits. The constructs were then washed gently with warm PDMS at 37 °C, followed by the secondary crosslinking using transglutaminase (5 U mL^{-1}) for 30 min.

Microelectrodes Electrochemical Characterization: Electrochemical characterization of the microelectrodes was performed using a Metrohm Autolab (PGSTAT302N) potentiostat in 1x PBS. An Ag/AgCl electrode (Sigma-Aldrich Z113085) was used as reference, and a Platinum foil was used as counter electrode. The microelectrodes were characterized by performing cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS) and charge injection capacity (CIC) determination. The CV was performed between –0.6 V and 0.8 V (electrochemical window of Pt in PBS against Ag/AgCl[47]) (6 cycles, 200 mV s^{-1}). The cathodic charge storage capacitance (CSCc) was calculated from CV measurements by integrating the cathodic of the measured current and normalizing by the scan rate. The EIS was measured between 1 Hz and 100 kHz. CV was always performed prior to EIS characterization. For the CIC determination, the electrodes were pulsed with a set current (for 300 µs at negative, 100 µs at 0, and 300 µs at positive) 10 times. The potential of the microelectrode was recorded, and this step was repeated for increasing currents. The current was ramped up until the electrode polarization (removing the ohmic drop) surpassed the electrochemical window.

Data analysis: Electrochemical characterization data were analyzed using Python 3.8 packages (Pandas, Numpy, Scipy, Lmfit, Matplotlib). Microelectrode charge injection capacity (CIC) was established by determining at what current the potential at the surface of the electrode (removing the ohmic drop) surpassed the water window (–0.6 V for cathodic pulses, and 0.8V for anodic pulses versus Ag/AgCl).

SEM: The SEM images of the electrodes after the aging tests were taken on a TFS Magellan 400 with a beam voltage of 5 kV and current of 100 pA. Prior to imaging, the samples stored in PBS were washed with DI water and dried. They were then sputtered with PtPd (6nm) to create a conductive layer (Safematic CCU-010 HV).

Mechanical Stability Evaluation—Adhesion test: The adhesion of Pt/Ti/SiO_2 tracks to the PDMS was tested performing a tape test. Standard scotch tape was laminated onto the MEA, pressed down by hand to ensure good contact, then peeled off. Optical images were taken before and after to assess the robustness of the tracks.

Stretching Test: A MEA was mounted on a homemade stretch setup consisting of two linear bearings actuated by a micrometer screw each. The device was clamped onto the bearings and the assembly was brought under an optical microscope (ADD model). The device was stretched manually using the micrometer screws and pictures were taken at regular intervals.

Spike Propagation Analysis on HD-MEAs—Mounting PDMS microstructures on HD-MEAs: HD-MEA chips were plasma activated for 1 min (air plasma) and coated with 0.1 mg mL^{-1} poly-D-lysine (PDL) in PBS for 30 min, washed three times with ultrapure water and dried using a nitrogen gun. Extracted PDMS microstructures were aligned on top of the sensing area and filled with matrigel through desiccation (1:25 in RGC medium on ice). The matrigel solution was immediately after replaced with 1 mL RGC or Neurobasal medium. Medium was changed every 3 days. Recordings were performed the day after changing the medium.

Spheroid Seeding: Primary retinal or cortical neurons were aggregated into spheroids as described in section “spheroid formation” at a size of 800 cells/spheroid and transfected with an EGFP expressing AAV (scAAV-DJ/2-hSyn1-chl-loxP-EGFP-loxP-SV40p(A)) at 15000 vg/cell. Two neural spheroids were manually seeded into each of the 16 seeding wells as described in section “spheroid generation”.

Acquisition of Spontaneous and Stimulation-Based Electrical Activity Recordings with HD-MEAs: For the spike propagation analysis within the biohybrid microstructures we used HD-MEAs (26400 electrodes arranged in a grid of 220 × 120 electrodes, 3.85 × 2.1 mm^2, pitch = 17.5 µm, MaxOne+, MaxWell Biosystems, Switzerland) with bare platinum electrodes and a flat surface topology. A maximum of 1024 stimulation electrodes can be simultaneously connected to the amplifiers for recording. We recorded from about 1000 electrodes specifically selected to cover the whole biohybrid design spanning area. Additionally, we established regions of interest at the exit of each seeding well and along the nerve channel for the stimulation-induced propagation analysis. Stimulation was performed with a single electrode. Axons inside each channel were sequentially stimulated with a rectangular cathodic-first biphasic pulse at 2 V peak-to-peak, 4 Hz and 400 µs pulse width. Data were collected at a sampling frequency of 20 kHz, with a 10-bit resolution and a recording range of about 3.2 mV, yielding a least significant bit (LSB) of 6.3 µV. Using customized software developed with the MaxWell Python API, we stored the raw traces S14 and performed spike detection with a custom algorithm.[62]

Data Analysis of Spontaneous and Stimulation-Based Electrical Activity recordings: Raw voltages traces were band-passed filtered (4th order acausal Butterworth filter, 300-3500 Hz). The baseline noise of the signal was characterized for each electrode using the median absolute deviation (MAD). Spikes were detected by identifying negative signal peaks below a threshold of five times the baseline noise. Successive events within 2 ms were discarded to avoid multiple detection of the same spike. The response upon electrical stimulation consisted in detecting spikes occurring within a time window of 3.5 ms after each stimulation pulse at each specific region of interest characterized by a set of electrodes (set of 5 electrodes at input channels Ch1-16 and set of 15 electrodes at target regions 1-3). Quantification of the response was done by binning the detected spikes with a bin size of 0.5 ms for each individual region. The response probability was calculated as the normalized number of detected spikes in each recorded channel or region after stimulation. The total expected response is the number of stimulation pulses divided by the number of selected recording electrodes at the regions). The latency maps were obtained by calculating the stimulation-triggered average spike waveform for all recorded electrodes (time window of 3.5 ms). Latency values correspond to the time of the maximum absolute amplitude of the obtained average spike waveform in this time window. For clarity, latency is only plotted for electrodes that exhibit an average spike waveform amplitude above half of the maximum waveform amplitude across all electrodes.

Fluorescence Imaging of the Networks on HD-MEAs: For fluorescence imaging of the CMOS chips using an inverted confocal microscope (Olympus, FluoView 3000, Lasers: 488nm, 561nm) all excess medium was removed and a round glass coverslip (10 mm diameter) was mounted on top of the PDMS microstructure. The surface tension between the cover slip and the chip enabled us to invert the chip for imaging in an inverted microscope. For mounting in the CLSM, the CMOS chip was placed in the recording unit which in turn was mounted into a custom made metal insert that fits into the stage of the microscope. This configuration enabled inverted mounting for imaging.

In Vitro Experiments—Fluorescent Imaging: All in vitro fluorescent imaging was performed in a Olympus FluoView 3000 equipped with a cell incubation chamber with temperature and CO_2 control. The lasers used were either 488nm or 561nm depending on the AAV used (respectively mRuby or EGFP). Image processing was carried out with Fiji (ImageJ).

Calcium Activity Imaging: For calcium activity experiments, devices and spheroid were prepared as described above and transfected with GCaMP8m. A picture of the setup can be seen in Figure S15a. Image acquisition was performed in a CLSM (Olympus, FluoView 3000, Laser: 488nm) using the resonant scanner and an acquisition rate of 10 images per second. Post processing of the movies was performed in Fiji (ImageJ).

In Vivo Experiments—Animals: Animal experiments were performed in accordance with standard ethical guidelines (European Communities Guidelines on the Care and Use of Laboratory Animals, 86/609/EEC) and were approved by the Veterinary Department of the Canton of Basel-Stadt. For all in vivo experiments, both male and female 30–120 days old wild-type (WT) mice of C57BL/6 background were used, maintained on a normal 12-hour light/dark cycle, and group-housed in a pathogen-free environment with ad libitum access to food and drinking water.

Preparation of the Biohybrid Implant for In Vivo Experiments: For in vivo implantation, the implants were fabricated and assembled as described in section “Biohybrid Implant Fabrication.” The implants were seeded with neurons either 7, 1 or 0 days before implantation. Implants had to be transferred from ETH Zurich to the IOB in Basel. For transport, the retinal ganglion cell medium was replaced with retinal ganglion cell medium containing 1 % methylcellulose. The resulting increased viscosity reduced medium turbulence during transport and prevented neural spheroids from being washed out of the seeding wells. For transport, a custom-made battery powered 5 % CO_2 and 37 °C temperature controlled incubation box “inkugo”[63] was used. Implants were stored inside inkugo until implantation.

Surgical Procedures and Implantation of the Biohybrid Implant: Surgeries were performed in 30-60 day old WT mice. Prior to the start of surgery, animals were anaesthetized using a Fentanyl/Medetomidine/Midazolam (FMM) mixture (0.05 mg/kg Fentanyl, 0.5 mg/kg Medetomidine, 5 mg/kg Midazolam; subcutaneous injection). Hair were removed from the head using a trimmer (Isis GT421, Aesculap). Animals were then placed in a stereotaxic frame (Model 1900, KOPF Instruments) and ocular gel (Humigel, Virbac) was applied to prevent dehydration of the cornea during surgery. A local anaesthetic (0.25 % Bupivacaine, 0.2 % Lidocaine) was injected at the site of skin incision several minutes before removing the skin using fine surgical scissors. The exposed skull was then carefully cleaned and cleared of connective tissue before being stereotaxically aligned using the anatomical bregma and lambda coordinates. A custom-made stainless steel head-mounting bar was fixed onto the skull using dental cement (Superbond C&B). After the cement hardened, a 4-mm diameter craniotomy was made over the left hemisphere with a dental drill, exposing the visual cortex and the cortical surface above the location (from skull at bregma, medial-lateral: -2.0 mm, anterior-posterior: -2.2 mm) of the dorsolateral geniculate nucleus (dLGN). Given that in most cases, devices were implanted epidurally, the dura mater was kept intact during the craniatomy. In case the device was implanted subdurally, the dura was carefully removed using a fine forceps. Using the stereotaxic setup, an injection needle (26G, 0.45 mm diameter) was then slowly inserted to a depth of 2.8 mm (from skull at bregma) to form a physical path to implant the device into the dLGN. The biohybrid device was then manually placed onto the cortex such that the neuronal seeding well openings faced the cortical surface allowing for direct nutrient support for the on-chip grown neurons. The nerve forming PDMS channel or collagen tube were inserted into the brain using the prepared physical path until the microstructure of the device aligned onto the cortical surface. A small round cover glass of 4-mm diameter (CS-4R, Warner Instruments) was placed on top of the biohybrid device and while gently applying pressure to seal the craniatomy and flatten the surface of the brain, it was carefully fixed to the skull using a cyanoacrylate adhesive (Pattex Ultra Gel). Finally, the remaining exposed parts of the device including the connector were covered in light-curing dental cement (ESPE RelyX Unicem 2, 3M) to increase their robustness for in vivo applications. At the end of the surgery, a Wake mix solution (0.5 mg kg^{-1} Flumazenil, 2.5 mg kg^{-1} Atipamezol; subcutaneous injection), was administered for recovery and Buprenorphine (0.1 mg kg^{-1}; subcutaneous injection) was applied to reduce postoperative pain. Analgesia was maintained by Carprofen (4 mg kg^{-1}; subcutaneous injection every 12 h) for 48 h.

In Vivo Two-Photon Imaging of Axon Growth in the Biohybrid Implant: Axon growth in the biohybrid implant was assessed in vivo using a two-photon microscope (FEMTOSmart series, Femtonics). For day 0 imaging, mice were head-fixed under the microscope just after the implantation surgery, prior to the end of anaesthesia, and placed onto a heating blanket. For imaging sessions on later days, recordings were performed in awake mice, which were first habituated to head fixation on the imaging setup while being able to move on a horizontal running wheel similar to the one in their home cage. Two-photon images were acquired with a 16X water immersion objective (Nikon, NA = 0.8) and the emitted fluorescence was detected by photomultiplier tube detectors (PMTs; high sensitivity GaAsP detectors, Femtonics). mRuby was detected by a red PMT and autofluorescence was simultaneously imaged by means of a green PMT. The MESc software (Femtonics) was used to control the microscope in resonant scanning mode and the laser (InSight X3, Spectra-Physics), which was tuned to either 980 nm or 1045 nm. A transparent ultrasound gel (GEL G008, FIAB) was placed on top of the glass window in which the objective was immersed for imaging. Imaging positions in the horizontal plane were varied by moving the mouse together with head-fixation device, which was mounted on a motorized stage (MT-1078/MT-2078/MT-2278, Sutter).

In Vivo Two-Photon Calcium Imaging of Neurons in the Biohybrid Implant: To record the calcium activity of neurons in the biohybrid implant, the same two-photon microscope as described in the previous section was used. The spontaneous calcium activity of the GCaMP8m^{+} neurons was assessed by head-fixing the mouse under the microscope while it was able to move on the horizontal running wheel. GCaMP8m was excited at a wavelength of 930 nm and frames were recorded at 26.8 Hz. Spontaneous activity in each imaging location was recorded for about 4–6 min.

Data Analysis of Calcium Activity Recordings: Calcium activity recordings were analysed by custom-written routines in MATLAB (Mathworks). Given that recordings were performed in awake animals, horizontal motion in the imaging field was first corrected using the NoRMCorre algorithm.[64] Then, cells were manually selected on the mean projection image of the corrected calcium imaging movie using the region of interest (ROI) plugin in ImageJ (https://imagej.net). The calcium traces of a single cell were computed by taking the average fluorescence within the ROI at any given timepoint. Additionally, for each cell, to correct for non-specific neuropil fluorescence contaminating the ROI, the 30th percentile of the fluorescence within a 10-pixel wide ring at a 5-pixel distance from the ROI edge was subtracted from the mean ROI fluorescence, as it is commonly done in calcium imaging.[65] It was also ensured that the neuropil rings did not contain other cells. Finally, the calcium trace of each cell was subtracted and divided by a slow (20-s long) moving low-pass (30th percentile) filtered version of itself to obtain a normalize $\Delta F/F$ trace. The final $\Delta F/F$ trace was then filtered by a 200-ms long average filter for display purposes.

Histology and Immunochemistry: At the end of experiments mice were deeply anaesthetized with a Ketamine/Xylazine mixture (120 mg kg^{-1} Ketamine, 16 mg kg^{-1} Xylazine; intraperitoneal injection) and transcardially perfused with cold phosphate buffered saline (PBS) followed by 4 % paraformaldehyde (PFA) in PBS. Implants were removed and brains were extracted and placed in 4 % PFA for storage at 4 °C. To confirm the correct targeting of the implant within the neural tissue, brains were washed three times for 10 min in PBS, embedded in 4 % agarose and cut at a thickness of 150 µm using a vibratome (VT1000S vibratome, Leica Biosystems). Brain slices were stained with Hoechst 33342 (10 µg/mL) in PBS and then embedded in ProLong Gold Antifade Mountant (Thermo Fisher Scientific). Slices were imaged around the implantation site using a confocal microscope (Olympus, FluoView 3000, Lasers: 405 and 561 nm).

For staining of the biohybrid implants at the end of the experiment, isolated implants were incubated for 2 h in blocking buffer containing 3% (v/v) normal donkey serum (NDS) (S30-100ML, Merck), 0.5% (v/v) Triton X-100 (93443-100ML, Sigma) and 1% (w/v) BSA (A4161-1G, Sigma) in PBS. A rat anti-RFP monoclonal primary antibody (for mRuby, 5F8, ChromoTek, 1:500), a chicken anti-GFAP primary polyclonal antibody (ab4674, Abcam, 1:500) and a rabbit anti-IBA1 primary monoclonal antibody (ab178846, Abcam, 1:500) were then added to the implants in the same buffer solution and incubated for 2 days at 4 °C. Alexa Fluor 488 Goat anti-Chicken IgG (H+L) secondary antibody (Invitrogen, A11039, 1:1000), Alexa Fluor 568 Donkey anti-Rat IgG (H+L) secondary antibody (Invitrogen, A78946, 1:1000) and Alexa Fluor 647 Donkey anti-Rabbit IgG (H+L) secondary antibody (Invitrogen, A31573, 1:1000) incubation were subsequently performed for 2h at room temperature in the type of buffer solution as before and supplemented with Hoechst 33342 (10 µg mL^{-1}). Implants were imaged using a confocal microscope (Olympus, Olympus IXplore SpinSR), Lasers: 405, 488, 561, and 640 nm).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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L.S. and A.F. contributed equally to this work. T.R. conceived and designed the study. A.F. and T.R. performed in vivo experiments. L.S. and T.R. designed the biohybrid implants. L.S. fabricated and characterized the biohybrid implants. B.F.C. performed stimulation experiments on CMOS and analyzed spike propagation on CMOS MEAs. L.S. and T.R. performed and analyzed in vitro stimulation experiments. A.B. characterized axonal growth in glass tubes. T.R. characterized axonal growth in collagen conduits. S.M. provided collagen conduits. S.S. and S.J.I. optimized directionality of axonal growth. E.C., J.D., C.M.T, T.R., and L.S. optimized coating conditions for axonal growth on PDMS. L.M. and T.R. characterized axonal growth in the final implants. P.C. fabricated GelSH-GelNB conduits.

L.S. and J.H. performed SEM imaging. S.J.I. designed the implant PCBs. K.V. contributed to neural spheroid preparation. J.V., B.R., S.W., S.J.I, B.M., C.M.T., K.V., and S.M. provided advice in various aspects of the project. The manuscript was jointly written by L.S., A.F., B.F.C., and T.R. and was revised by all authors.

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Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

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